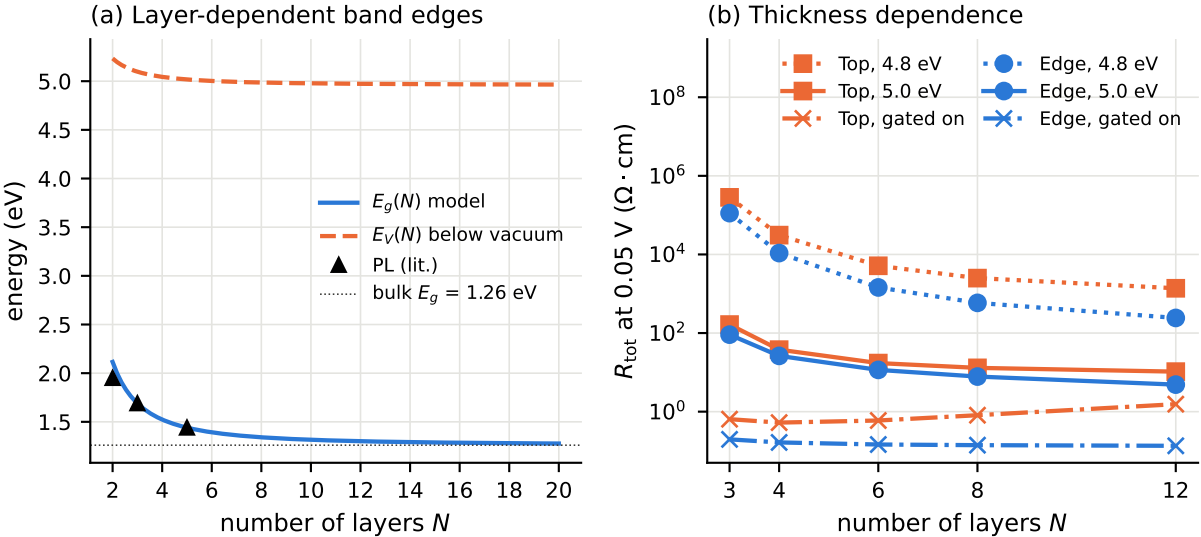
(a) Layer-dependent band edges
energy (eV)
5.0
4.5
4.0
3.5
3.0
2.5
2.0
1.5
$E_g(N)$ model
$E_V(N)$ below vacuum
PL (lit.)
bulk E_g = 1.26 eV
2
4
6
8
10
12
14
16
18
20
number of layers N
(b) Thickness dependence
R_{tot} at 0.05 V ($\Omega \cdot$ cm)
10^8
10^6
10^4
10^2
10^0
Top, 4.8 eV
Top, 5.0 eV
Top, gated on
Edge, 4.8 eV
Edge, 5.0 eV
Edge, gated on
3
4
6
8
12
number of layers N